

Kubernetes and Cloud Native Associate - KCNA

Kubernetes Resources

Demo Pods

In this tutorial, you'll learn how to deploy a pod in your Minikube cluster. A pod is the smallest and simplest deployable unit in Kubernetes, designed to hold one or more application containers. We'll use the `kubectl` command-line tool to interact with our cluster.



Note

You can specify an image tag or use an alternative container registry if your desired image is hosted elsewhere.

Pod Operations

Creating a Pod

To create a pod named "nginx" using the Docker image "nginx" (pulled from Docker Hub), run the following command:

```
kubectl run nginx --image=nginx
```

Once this command executes, Kubernetes creates the pod. You can verify its creation and status by checking the list of pods:

```
# kubectl run nginx --image=nginx
pod/nginx created

# kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
nginx	1/1	Running	0	3s

The output confirms that the pod is running. The "READY" column indicates how many containers are in a ready state, "RESTARTS" shows the number of times the container has restarted, and "AGE" reflects how long the pod has been active.

Inspecting Pod Details

For more in-depth information about the pod—including its labels, node assignment, internal IP address, and container specifics—use the ``kubectl describe`` command:

```
# kubectl run nginx --image=nginx
pod/nginx created

# kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
nginx	1/1	Running	0	3s

```
# kubectl describe pod nginx
```

Name: nginx
Namespace: default
Priority: 0
Node: minikube/192.168.99.100
Start Time: Sat, 11 Jul 2020 00:49:39 -0400
Labels: run=nginx
Annotations: <none>
Status: Running
IP: 172.17.0.3
IPs:
IP: 172.17.0.3

Containers:**nginx:**

```

Container ID:   docker://987785b312ad2e38c77132300f8709b8a027566462c2d18634ff:
Image:          nginx
Image ID:       docker-pullable://nginx@sha256:a23a9056789b968a186c5205f4

```

This command output provides essential metadata and status details such as the pod's start time, the node it is running on, and its internal IP address (172.17.0.3). If multiple containers were running within the pod, each would be listed under the "Containers" section.

Additionally, the bottom section of the output displays event information that tracks the lifecycle of the pod—from scheduling on the Minikube node to pulling the image and finally creating and launching the container:

```
Initialized           True
```

```
Ready                 True
```

```
ContainersReady       True
```

```
PodScheduled          True
```

Volumes:**default-token-f5ntk:**

```
Type:                Secret (a volume populated by a Secret)
```

```
SecretName:          default-token-f5ntk
```

```
Optional:            false
```

```
QoS Class:           BestEffort
```

```
Node-Selectors:      <none>
```

Tolerations:

```
node.kubernetes.io/not-ready:NoExecute for 300s
```

```
node.kubernetes.io/unreachable:NoExecute for 300s
```

Events:

Type	Reason	Age	From	Message
----	-----	----	----	-----
Normal	Scheduled	46s	default-scheduler	Successfully assigned defa
Normal	Pulling	45s	kubelet, minikube	Pulling image "nginx"
Normal	Pulled	44s	kubelet, minikube	Successfully pulled image
Normal	Created	44s	kubelet, minikube	Created container nginx
Normal	Started	44s	kubelet, minikube	Started container nginx

Viewing Pod Information in Wide Format

For a summary that also includes node and internal IP details, use the following command:

```
kubectl get pods -o wide
```

The output will resemble:

NAME	READY	STATUS	RESTARTS	AGE	IP	NODE	NOMINATED NO
nginx	1/1	Running	0	2m28s	172.17.0.3	minikube	<none>

Here, each pod is assigned its own internal IP address (in this case, 172.17.0.3), which enables network communications within the cluster.

This demonstration has shown how to deploy a pod in a Minikube environment using the `kubectl` command line. In upcoming lessons, we will explore how to define pods using YAML configuration files for more complex deployment scenarios.

For further reading and detailed Kubernetes concepts, visit the [Kubernetes Documentation](#).



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